

Decentralized Iris-Boosted Fuzzy Vault (D-IBFV)

Zero-Penalty Multimodal Biometric Template Protection with Threshold-Distributed Storage

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Motivation

Background and Related Work

Contribution 1: IBFV

Contribution 2: D-IBFV

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Motivation

- ▶ **Aadhaar (India, 2018):** 1.1 billion biometric records exposed in a single breach of the centralised UIDAI database
- ▶ **US Office of Personnel Management (2015):** 5.6 million fingerprint records stolen; used for intelligence operations
- ▶ **BioStar 2 (2019):** 28 million biometric records including fingerprints and facial templates left exposed on an unprotected server
- ▶ **India's FASTag (2023):** facial and fingerprint data of millions of vehicle owners leaked

Why biometrics are uniquely dangerous:

- ▶ A stolen password can be changed in seconds
- ▶ A stolen fingerprint **cannot be changed — ever**
- ▶ An attacker with your biometric template has lifelong leverage
- ▶ Every new system you enroll in is instantly compromised

Raw template storage (status quo):

- ▶ Biometric scanner extracts a feature vector (minutiae, iris code)
- ▶ This raw vector is stored in a central database
- ▶ Attacker who steals the database can:
 - ▷ Replay the template to spoof any system that enrolled from it
 - ▷ Cross-match across databases to de-anonymize users
 - ▷ Reconstruct the physical trait using generative models (GAN attacks)

The single-server problem:

- ▶ Even protected templates stored on one server create one target
- ▶ A single database breach exposes the complete vault to offline attack
- ▶ Offline polynomial reconstruction: $\approx 6.8 \times 10^8$ attempts at $10^6/\text{sec} = \mathbf{11 \text{ minutes}}$ (detailed in Security section)
- ▶ Distributed storage eliminates this single point of failure

The international standard for biometric information protection defines three **mandatory** properties:

✓ **Irreversibility**

It must be computationally infeasible to reconstruct the original biometric trait from the stored template, even with full knowledge of the protection scheme and its parameters.

Why it matters: prevents reconstruction attacks; recovering the fingerprint from a stolen vault must require solving ECDLP (2^{128} operations).

✓ **Unlinkability**

Templates created from the same biometric for different applications must be mathematically unrelated. An adversary who compromises one application's database must learn nothing about the same user's template in another.

Why it matters: prevents cross-application de-anonymization.

✓ **Revocability**

If a template is compromised, it must be possible to cancel it and issue a new, independent template from the *same* biometric.

Why it matters: enables damage control after a breach.

No prior multimodal fuzzy vault scheme satisfies all three properties while also providing a formal zero-penalty multimodal fusion guarantee. This thesis does.

Background and Related Work

Core idea:

- ▶ Bind a secret s to a noisy, unordered set A (e.g., fingerprint minutiae coordinates)
- ▶ Encode s as a degree- k polynomial $p(x)$ over $\text{GF}(q)$
- ▶ Evaluate p at each element of $A \rightarrow$ genuine points $\{(a_i, p(a_i))\}$
- ▶ Add hundreds of **chaff points** (c_j, r_j) where $r_j \neq p(c_j)$ — these are fake, indistinguishable
- ▶ **Vault** = shuffle of genuine + chaff

Unlocking:

- ▶ Query features B are matched against vault x-coordinates
- ▶ If $|A \cap B| \geq k+1$ genuine matches found: Lagrange interpolation reconstructs $p(x)$ and recovers s
- ▶ Chaff points produce garbage polynomials — easily detected

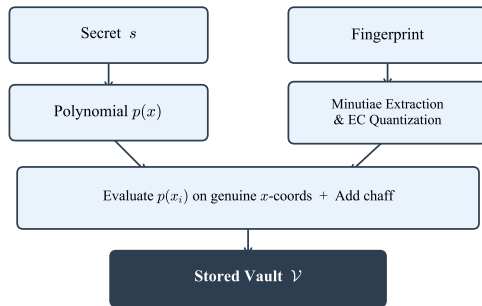
Security intuition:

- ▶ Without knowing A , attacker must try all $\binom{N}{k+1}$ subsets
- ▶ With N total points and most being chaff: brute-force is computationally infeasible
- ▶ Weakness: $\text{GF}(2^{16})$ gives only 2^8 security — too weak

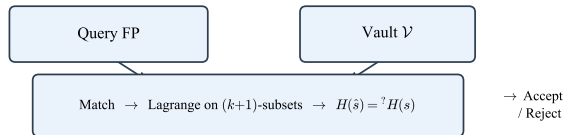
Why ECC? $\text{GF}(2^{16})$ gives only 2^8 security. ECC Brainpool P256r1 gives 2^{128} . Enrollment steps:

1. **Quantize** minutia (x_i, y_i, θ_i) by factor f : each dimension modulo 64 $\rightarrow$ 18-bit integer q_i
2. **EC Map**: $P_i = q_i \cdot G$ on Brainpool P256r1. Recovering q_i from $P_i \cdot x$ requires solving ECDLP: $\approx 2^{128}$ operations
3. **Polynomial**: degree- k poly $p(x)$ over $\text{GF}(p)$, coefficients in $[10, 63]$. Secret $s = \bigoplus \text{bin}_6(a_j) \bmod n$; public key $S = s \cdot G$
4. **Genuine points**: $\mathcal{G} = \{(P_i \cdot x, p(P_i \cdot x))\}_{i=1}^g$
5. **Chaff**: $5g$ random pairs $(c_j \cdot x, r_j)$ with $r_j \neq p(c_j \cdot x)$
6. **Vault**: $\mathcal{V} = \text{shuffle}(\mathcal{G} \cup \mathcal{C})$; stored with public key S

(a) Vault Locking



(b) Vault Unlocking



Given a query fingerprint, unlock the vault:

1. Quantize query minutiae $\mathcal{M}'$ to $\mathcal{Q}'$; compute $\{P'_i = q'_i \cdot G\}$
2. Match query EC points to vault x-coordinates: $\mathcal{X}_m = \{(x, y) \in \mathcal{V} : x \in \{P'_i.x\}\}$
3. If $|\mathcal{X}_m| \geq k+1$: try each $(k+1)$ -subset via Lagrange interpolation, recover candidate $\hat{s}$
4. Verify: accept if $\hat{s} \cdot G = S$

Performance of baseline (Maurya 2025):

Database	DB1	DB2	DB3	DB4
Best EER (%)	0.000	0.000	2.762	2.404

- DB1 and DB2 (high quality): perfect accuracy
- DB3 (thermal sensor, poor quality) and DB4 (deliberate distortion): **still non-trivial EER even with best-in-class ECC vault**
- This is where iris-boosting helps most

Why iris? Daugman (2004) showed iris has the widest intra/inter-class separation of any biometric — FHD near 0.10–0.15 (genuine) vs 0.46 (impostor). **Processing pipeline:**

1. **Capture:** CASIA-Iris-Interval, 320×280 px, near-infrared illumination
2. **Segment:** Detect pupil (intensity thresholding) and outer iris (radial gradient). Mask pupil, specular reflections, eyelids
3. **Normalize:** Daugman rubber-sheet model $\rightarrow 64 \times 512$ rectangular strip
4. **Encode:** CLAHE enhancement, 1-D Log-Gabor filters (18 and 36 px wavelengths) $\rightarrow$ **131,072-bit iris code** (phase-quantized binary)
5. **Match:** Fractional Hamming Distance (FHD) over valid bits, ± 32 column shifts for rotational tolerance

Key property for fusion:

- ▶ Genuine FHD ≈ 0.10 – 0.15 (low = similar)
- ▶ Impostor FHD ≈ 0.46 (close to random = 0.5)
- ▶ Wide separation enables reliable binary key extraction

Goal: extract a stable, secret binary key from a noisy iris code. **Enrollment:**

1. Partition 131,072-bit code into n_b blocks using **bit interleaving**: bit j goes to block $(j \bmod n_b)$. This *spreads* spatially correlated noise (eyelid occlusion) across *all* blocks instead of wiping out consecutive ones
2. Generate random n_b -bit key $\mathbf{k}$
3. Helper data: $\mathbf{h} = \mathbf{c} \oplus \mathbf{w}$ (where $\mathbf{w}$ repeats $\mathbf{k}$ over all positions)
4. Store $(\mathbf{h}, H(\mathbf{k}))$ — *not* $\mathbf{k}$ itself

Recovery (at query time):

1. For each rotation shift $\delta \in [-32, +32]$: compute $\tilde{\mathbf{c}} = \text{rotate}(\mathbf{c}', \delta)$, then $\tilde{\mathbf{w}} = \tilde{\mathbf{c}} \oplus \mathbf{h}$
2. For each block: **majority vote** over block's valid bits $\rightarrow \hat{k}_i$
3. At $B_s = 255$: capacity = 127 errors/block ($\approx 49.8\%$); genuine FHD ≈ 0.20 (≈ 51 errors/block) — majority vote succeeds
4. Accept if $H(\hat{\mathbf{k}}) = H(\mathbf{k})$

Block size B_s	Key bits n_b	GAR	Impostor FAR
255	514	$\approx 62\%$	$\approx 0.10\%$

What it does:

- ▶ Include genuine points from *both* fingerprint and iris in the same vault
- ▶ Polynomial reconstruction requires $\geq k+1$ matches drawn from *either* modality
- ▶ But in practice: the **AND constraint** applies — both modalities must contribute enough points

The gating penalty:

$$\text{GAR}_{\text{AND}} = \text{GAR}_{\text{FP}} \times \text{GAR}_{\text{Iris}} \leq \text{GAR}_{\text{FP}}$$

Our iris GAR $\approx 62\%$. So **38% of genuine users are blocked by iris alone**, even when their fingerprint is perfect.

EER results for Architecture A vs Unimodal:

	DB1	DB2	DB3	DB4
Unimodal	0.000	0.000	2.762	2.404
Arch. A (AND)	1.000	1.000	3.793	3.045

AND-fusion is **worse** than unimodal on every database. Nandakumar & Jain (2008) and Moon et al. (2010) both suffer this fate.

Three alternative tight-coupling strategies all make the iris a **mandatory decryption key**: **Architecture B** —

Polynomial Blinding:

- ▶ XOR-mask each polynomial coefficient with an iris-derived key
- ▶ Blind polynomial $\rightarrow$ vault; recover key $\rightarrow$ unmask $\rightarrow$ decrypt
- ▶ Without iris key: polynomial is permanently masked. **Vault is useless.**

Architecture C — Dense Chaff Seeding:

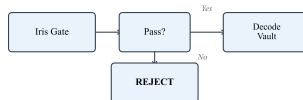
- ▶ Iris key seeds a PRNG that injects $20\times$ extra chaff
- ▶ Without iris key: search space inflates $20\times \rightarrow$ computationally infeasible to reconstruct
- ▶ **User is rejected even if fingerprint matches perfectly.**

Architecture D — AES-256-GCM Vault Encryption:

- ▶ Entire vault is AES-encrypted with the iris-derived key
- ▶ Without iris key: vault is a block of ciphertext. **Cannot even begin to decode.**

All four architectures share one structural flaw:

(a) AND-Fusion (Gating)



$$GAR_{AND} = GAR_{FP} \times GAR_{Iris} \leq GAR_{FP}$$

(b) IBFV (Boosting)



Architecture	DB1	DB2	DB3	DB4	Iris mandatory?
Unimodal	0.000	0.000	2.762	2.404	No
A (AND)	1.000	1.000	3.793	3.045	Yes
B, C, D (gate)	19.00	19.00	19.47	19.00	Yes (hard)

B, C, D: iris $GAR \approx 62\% \Rightarrow FRR \geq 38\%$ by construction. EER floor $\approx 19\%$ — **no parameter tuning can fix this**. The problem is architectural, not numerical. **What we need:** a design where iris *helps when available and causes zero harm when*

The mathematics:

- ▶ Architectures B, C, D require iris key for vault decoding
- ▶ Iris key recovery GAR $\approx 62\%$
- ▶ Therefore: $FRR \geq 1 - 0.62 = 38\%$ by construction, *regardless of fingerprint quality, threshold, or parameters*

EER calculation:

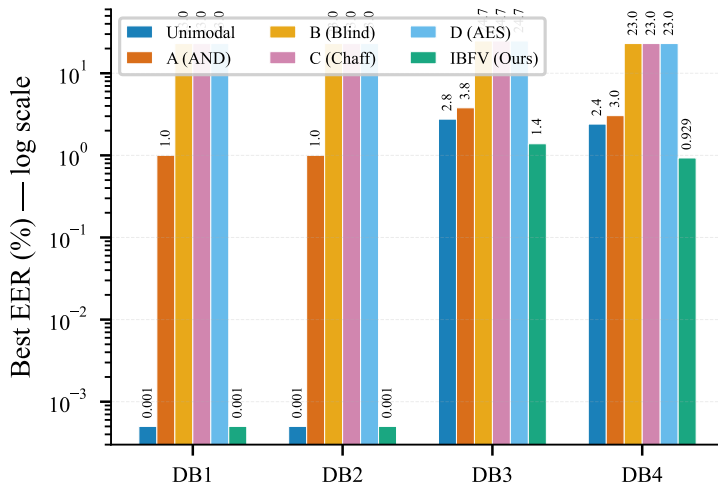
$$EER = \frac{FRR + FAR}{2} \approx \frac{38\% + 0\%}{2} = 19\%$$

- ▶ This floor is **structural** — it cannot be removed without changing the architecture
- ▶ Any tuning that raises FAR to meet FRR at 19% confirms the floor
- ▶ The *only* solution: decouple iris from the decoding path

Setup 1 EER for B/C/D across all databases:

Architecture	DB1	DB2	DB3	DB4
Arch. B (Blinding)	19.00	19.00	19.28	19.00
Arch. C (Chaff)	19.00	19.00	19.47	19.00
Arch. D (AES)	19.00	19.00	19.47	19.00

Architecture Comparison — All Four vs Unimodal



Every prior multimodal architecture is **worse** than using fingerprint alone. The gap widens on harder databases (DB3, DB4). IBFV breaks this pattern.

Contribution 1: IBFV

The problem with all prior work:

They asked: "*How do we force iris into the decoding path?*"

This created the mandatory gate and the 19% floor. **The right question:**

"How can iris help when it works, without hurting when it does not?" **Our answer — Iris Bonus Points:**

- ▶ When iris succeeds ($\sim 62\%$ of genuine queries): derive K virtual polynomial evaluation points from the iris key
- ▶ Inject these as **guaranteed genuine points** into the vault's candidate set
- ▶ Smart decoder fixes all K bonus points, reducing the search space by $\approx 5.5\times$
- ▶ More genuine points in the candidate set $\rightarrow$ higher chance of finding $k+1$ matches $\rightarrow$ lower EER
- ▶ When iris fails ($\sim 38\%$): **nothing changes** — fall back to unimodal exactly

The consequence:

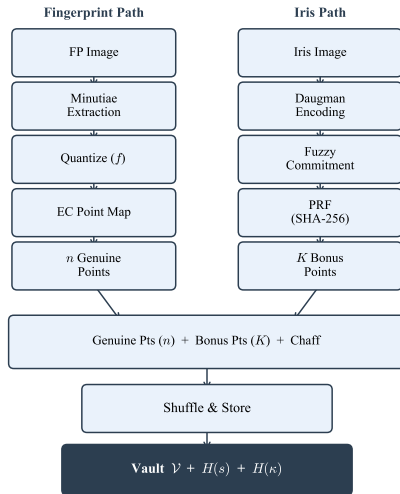
It is *mathematically impossible* for IBFV to perform worse than unimodal. Formally proven in Theorem 1.

After standard ECC vault enrollment, three additional steps:

1. Extract iris key $\mathbf{k}$; compute $\kappa = H(\mathbf{k})$
2. Derive K virtual x-coordinates via PRF (SHA-256): $v_j = \text{SHA-256}(\text{"ibfv_vpt:"} \parallel j \parallel \kappa) \bmod p$
3. Compute bonus genuine points: $\mathcal{G}_{\text{Iris}} = \{(v_j, p(v_j))\}_{j=0}^{K-1}$
4. Final vault: $\mathcal{V} = \text{shuffle}(\mathcal{G}_{\text{FP}} \cup \mathcal{G}_{\text{Iris}} \cup \mathcal{C})$

Without κ , the K x-coordinates are computationally indistinguishable from chaff (SHA-256 preimage resistance).

IBFV Enrollment



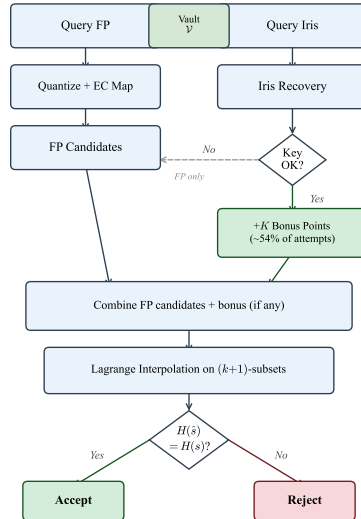
When iris key recovery succeeds:

1. Recover iris key $\hat{\mathbf{k}} \rightarrow$ derive $v_0, \dots, v_{K-1}$ (same PRF as enrollment)
2. Locate bonus matches in vault: $\mathcal{X}_{\text{Iris}} = \{(x, y) \in \mathcal{V} : x \in \{v_j\}\}$
3. Total candidate set: $\mathcal{X} = \mathcal{X}_{\text{FP}} \cup \mathcal{X}_{\text{Iris}}$ (strictly larger than unimodal)
4. **Smart decoding:** fix all K bonus points (guaranteed genuine by construction), search only $k+1-K$ remaining from $\mathcal{X}_{\text{FP}}$

Search space reduction example ($k = 10$, $K = 4$, $|\mathcal{X}_{\text{FP}}| = 12$):

$$\frac{\binom{16}{11}}{\binom{12}{7}} = \frac{4368}{792} \approx 5.5 \times \text{fewer combinations to check}$$

IBFV Verification



When iris key recovery fails:

- ▶ Set $\mathcal{X} = \mathcal{X}_{\text{FP}}$ only
- ▶ Proceed with standard Lagrange interpolation — **identical to the unimodal vault in every way**
- ▶ Bonus points in the vault are indistinguishable from chaff (attacker cannot use them; verifier does not use them)
- ▶ **Zero change. Zero penalty. Zero degradation.**

This is the key innovation:

In all prior multimodal vaults, iris failure means vault failure.

In IBFV, iris failure means unimodal operation — nothing worse. **Contrast with all prior architectures:**

Architecture	Iris fails $\Rightarrow$	Effect
Arch. A (AND)	Half the genuine set gone	EER rises
Arch. B (Blind)	Polynomial masked	Vault unusable
Arch. C (Chaff)	20× chaff not stripped	Vault unusable
Arch. D (AES)	Cannot decrypt	Vault unusable
IBFV	Fall back to unimodal	Zero change

Theorem 1 (Zero-Penalty Guarantee). *For any quantization factor f , polynomial degree k , and any biometric sample pair $(\mathcal{M}', \mathbf{c}')$:*

$$\Pr[\text{Accept}_{\text{IBFV}}] \geq \Pr[\text{Accept}_{\text{Uni}}]$$

Proof (two exhaustive cases):

► **Case 1 — Iris succeeds:**

$$\mathcal{X}_{\text{IBFV}} = \mathcal{X}_{\text{FP}} \cup \mathcal{X}_{\text{Iris}} \supseteq \mathcal{X}_{\text{FP}}$$

Every subset available to the unimodal decoder is also available to IBFV, plus additional genuine subsets.

$$\text{Accept}_{\text{Uni}} \Rightarrow \text{Accept}_{\text{IBFV}}.$$

► **Case 2 — Iris fails:**

$$\mathcal{X}_{\text{IBFV}} = \mathcal{X}_{\text{FP}} \text{ — identical to unimodal in all respects. (QED)}$$

Corollary: $\text{FRR}_{\text{IBFV}} \leq \text{FRR}_{\text{Uni}}$ for all parameter settings. Since iris impostor FAR $\approx 0.10\%$:

$\text{EER}_{\text{IBFV}} \leq \text{EER}_{\text{Uni}}$. **Empirical confirmation:**

- Setup 2: **zero violations** in all 144 tested configurations
- No prior multimodal fuzzy vault scheme has this formal guarantee

Contribution 2: D-IBFV

The centralised vault attack:

- ▶ The IBFV vault is a high-value target: it contains all the information needed for offline decoding
- ▶ Server compromise → attacker gets the complete vault → offline brute-force with no rate limiting
- ▶ Expected attempts for our vault (124 points, 24 genuine, $k=10$):

$$\frac{\binom{124}{11}}{\binom{24}{11}} \approx 6.8 \times 10^8$$

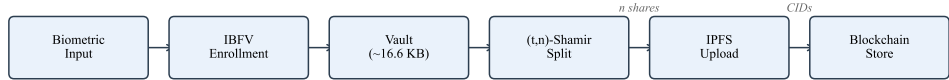
At 10^6 attempts/sec: $\approx$ **680 seconds (~11 minutes)**

Threshold distribution eliminates this:

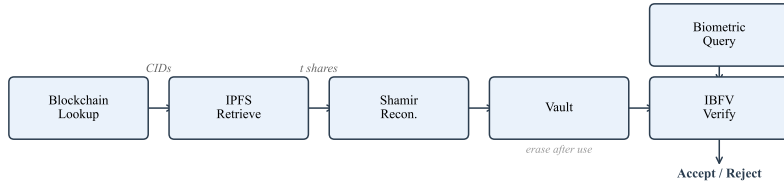
- ▶ Split vault into n shares using Shamir Secret Sharing
- ▶ Each share alone reveals **zero information** (information-theoretic, not just computational)
- ▶ Any t of n shares reconstruct the vault exactly
- ▶ Store shares on IPFS (distributed, content-addressed)
- ▶ Record share locations (CIDs) on blockchain (tamper-evident)

Result: An attacker must compromise t independent IPFS nodes *and* the blockchain record simultaneously.

Enrollment



Verification



Top: Enrollment — IBFV vault → Shamir split → IPFS (shares) → Blockchain (CIDs). **Bottom:** Verification — Blockchain (lookup CIDs) → IPFS (fetch *t* shares) → Shamir reconstruct → biometric verify.

Setup: (t, n) -threshold sharing over GF(257)

- ▶ Applied **independently to each byte** b_i of the serialized vault **b**
- ▶ Why GF(257)? Smallest prime field containing all 256 byte values (0–255). No truncation or modular artefacts.

Splitting (enrollment):

1. For each byte b_i : sample random polynomial $f_i(x)$ of degree $t-1$ over GF(257) with $f_i(0) = b_i$
2. Compute n shares: $s_{j,i} = f_i(j) \bmod 257$, for $j = 1, \dots, n$
3. Share j collects $\{s_{j,0}, s_{j,1}, \dots, s_{j,L-1}\}$ for all L vault bytes

Reconstruction (verification):

1. Collect any t shares
2. For each byte position i : Lagrange interpolation at $x = 0$ over GF(257) $\rightarrow$ exact $\hat{b}_i = b_i$

 (t, n) configurations used:

(t, n)	Fault tolerance	Split (ms)	Recon (ms)	IPFS storage
(2, 3)	1 node	0.3	0.2	99 KB
(3, 5)	2 nodes	0.7	0.2	166 KB
(4, 7)	3 nodes	1.3	0.2	232 KB

Theorem 2 (Share Confidentiality). *For any $t' < t$ compromised shares with distinct indices, the conditional distribution of vault $\mathbf{b}$ is uniform over $\{0, \dots, 255\}^L$:*

$$H(\mathbf{b} \mid \mathbf{s}_{j_1}, \dots, \mathbf{s}_{j_{t'}}) = H(\mathbf{b})$$

*Fewer than t shares reveal **zero information** about the vault content.*

Theorem 3 (Exact Reconstruction). *Given any t shares with distinct indices, Lagrange interpolation over $GF(257)$ recovers $\hat{b}_i = b_i$ for all bytes i , yielding $\hat{\mathbf{b}} = \mathbf{b}$ with zero bit errors.*

Empirical confirmation — 30,000 reconstruction trials:

Metric	(3, 5)	(4, 7)	(5, 9)
Trials	10,000	10,000	10,000
Bit errors	0	0	0
SHA-256 match	100%	100%	100%
Below threshold reconstruction	0%	0%	0%

Sharp threshold: exactly t shares needed; $t-1$ reveals nothing; both theorems confirmed experimentally.

What is IPFS?

- ▶ A peer-to-peer storage network where each file is identified by its **Content Identifier (CID)**, computed as the SHA-256 hash of its content
- ▶ Retrieving by CID = verifying integrity automatically: any tampering changes the CID, making the share undetectable as modified
- ▶ No single server hosts the data — any node that *pins* a CID serves it

Why IPFS for vault shares:

- ▶ **Integrity:** CID is a cryptographic commitment — tampered share has a different CID, lookup fails
- ▶ **Decentralization:** shares spread across n IPFS nodes; no single node is a single point of failure
- ▶ **Content addressing:** the blockchain stores only 0.5 KB of CID pointers — not the vault itself (16.6 KB)
- ▶ **Retrievability:** any node that has pinned the share can respond to retrieval; no DNS, no IP dependency

D-IBFV usage: Each of the n Shamir shares is uploaded as a separate IPFS object. The n CIDs are anchored on-chain. Verification: fetch any t shares using their CIDs.

What is a smart contract?

- ▶ A program deployed on the blockchain with **immutable code** and **transparent state**
- ▶ Anyone can call it; execution is verifiable by all nodes; once deployed, no one (including the deployer) can modify it

D-IBFV Solidity contract (CID Registry):

- ▶ **Enroll**: store $(uid, [cid_1, \dots, cid_n], t, n)$
- ▶ **Lookup**: return the n CIDs for a given uid
- ▶ **Revoke**: mark a user's enrollment as invalidated
- ▶ On-chain storage: **0.5 KB per user** (CID records only; vault stays on IPFS)

Platforms evaluated:

- ▶ **Ganache**: local EVM sandbox, instant block mining, no real gas cost — used as controlled latency baseline
- ▶ **Ethereum L1**: $\approx 209,000$ gas per enrollment at 30 Gwei $\approx$ \$18.80
- ▶ **Polygon / Layer-2**: same Solidity contract, $\approx$ \$0.001 per enrollment

Ganache enrollment latency: Blockchain TX = 245.7 ms out of 363 ms total (IPFS upload = 116.4 ms; crypto = < 2 ms)

What is Hyperledger Fabric?

- ▶ A **permissioned** enterprise blockchain — only known, credentialed participants can join the network
- ▶ Suitable for institutional deployments (government ID, hospital, financial institution) where participants are known and regulated

Fabric architecture for D-IBFV:

- ▶ **Chaincode (Go):** the CID registry smart contract, same logic as Solidity version
- ▶ **Endorsement policy:** N -of- M peers must endorse before a transaction is committed
- ▶ **Raft consensus:** crash fault tolerant, 0.5-second block timeout; ordering service produces blocks every 500 ms
- ▶ **Execute-Order-Validate model:** transactions executed by endorsers first, then ordered and validated (prevents read-write conflicts at commit time)

Fabric vs Ethereum for biometric systems:

Property	Ganache/Ethereum	Hyperledger Fabric
Participation	Permissionless	Permissioned
Gas cost	Yes (L1: \$18)	No
Lookup TPS (4 workers)	25.1	48.1
Enrollment latency	363 ms	806 ms
Privacy	Public ledger	Channel-based

Experimental Results

Fingerprint databases (FVC standard):

Database	Subjects	Sensor	Note
FVC2002 DB1	100×8	Optical	Clean
FVC2002 DB2	100×8	Capacitive	Clean
FVC2002 DB3	100×8	Thermal	Hardest
FVC2004 DB1	100×8	Optical	Deliberate distortion

Iris: CASIA-Iris-Interval, 249 subjects, NIR, 99.8% extraction rate.

Chimeric DB: FVC subject i paired with CASIA iris i . Modalities are *statistically independent* — conservative evaluation. **Evaluation protocols:**

- ▶ **Setup 1:** Impression 1 $\rightarrow$ enroll; Impression 2 $\rightarrow$ query. 100 genuine / 9,900 impostor per database.
- ▶ **Setup 2:** Impression 1 $\rightarrow$ enroll; Impressions 2–8 $\rightarrow$ queries. 336 genuine / 15,792 impostor. Harder cross-impression variability.

Parameter sweep: $f \in \{14, 16, 18, 20, 22, 24, 26, 28, 30\}$, $k \in \{5, 7, 9, 11\}$, $K \in \{0, 1, 2, 4, 6, 8\} \rightarrow$ **864 total configurations** per database per setup.

Architecture	DB1	DB2	DB3	DB4
Unimodal	0.000	0.000	2.762	2.404
Arch. A (AND-fusion)	1.000	1.000	3.793	3.045
Arch. B (Blinding)	19.00	19.00	19.28	19.00
Arch. C (Dense Chaff)	19.00	19.00	19.47	19.00
Arch. D (AES-GCM)	19.00	19.00	19.47	19.00
IBFV (ours)	0.000	0.000	1.384	0.929

Key reductions (Setup 1):

- ▶ DB3: 2.762% $\rightarrow$ 1.384% — **49.9% relative reduction**
- ▶ DB4: 2.404% $\rightarrow$ 0.929% — **61.3% relative reduction**
- ▶ DB1, DB2: already perfect unimodal; IBFV maintains perfection
- ▶ Every competing multimodal architecture is **worse** than unimodal

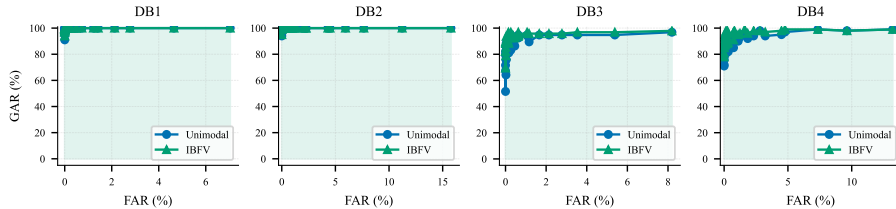
Architecture	DB1	DB2	DB3	DB4
Unimodal	2.736	4.588	6.738	4.307
Arch. A (AND-fusion)	3.264	5.143	6.741	4.800
IBFV (ours)	2.242	3.185	4.591	2.493

Key reductions (Setup 2):

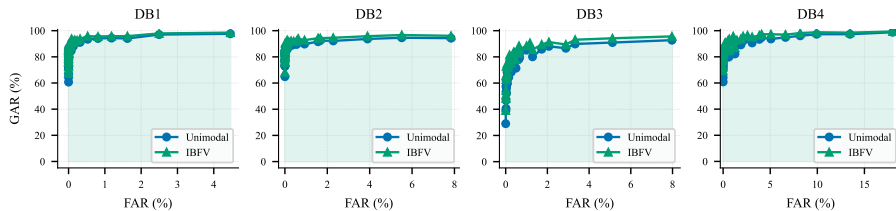
- ▶ DB3: 6.738% $\rightarrow$ 4.591% — **31.9% relative reduction**
- ▶ DB4: 4.307% $\rightarrow$ 2.493% — **42.1% relative reduction**
- ▶ DB1: 2.736% $\rightarrow$ 2.242% — **18.1% relative reduction**
- ▶ DB2: 4.588% $\rightarrow$ 3.185% — **30.6% relative reduction**

Critical: Theorem 1 zero-violations confirmed:

Across all **144 Setup-2 configurations** (4 DBs $\times$ 9 f values $\times$ 4 k values), IBFV **never exceeds unimodal EER**. Zero violations. The formal proof holds experimentally.

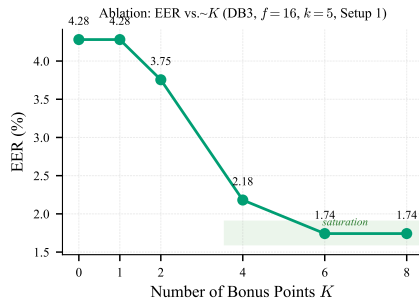


IBFV dominates unimodal at every operating point on all four databases. AUC improvement largest on DB3 and DB4.



Harder evaluation (7 impressions). IBFV still dominates at every operating point. DB3: AUC 0.9536 $\rightarrow$ 0.9756 (+2.20 pp).

Ablation — How Many Bonus Points K ?



K	EER (%)	Notes
0 (Unimodal)	4.281	Baseline
1	4.281	No benefit: one point cannot alone reach threshold
2	3.755	
4	2.181	First improvement 80% of maximum benefit; FAR elevation <0.16 pp
6	1.742	Saturation begins
8	1.742	Saturated: no further gain

Bootstrap 95% CI ($N=2000$): $K=4$ advantage over all alternatives: $p < 0.01$. Default $K=4$ throughout.

Metric: D_{sys} (Gomez-Barrero et al.): 0 = perfect unlinkability; 1 = fully linkable

Scenario	DB1	DB2	DB3	DB4
Same quantization factor (same app)	0.994	0.994	0.920	0.929
Different factor (different apps)	0.317	0.313	0.057	0.388
Diverse factor (recommended)	0.003	0.014	0.026	0.007

- ▶ Same factor (same application): linkable — expected and acceptable
- ▶ **Different factor per application** (recommended deployment): $D_{\text{sys}} \leq 0.03$ — **near-perfect unlinkability**
- ▶ Independent polynomial, chaff, and iris key per enrollment ensures mathematical independence across enrollments

Revocability: Re-enroll with new factor f' , new polynomial, new iris helper data $\Rightarrow$ entirely new vault domain. Old vault provides zero advantage (Theorem 2).

System Performance and Security

Theorem 3 empirical validation:

Metric	(3, 5)	(4, 7)	(5, 9)
Trials	10,000	10,000	10,000
Bit errors	0	0	0
SHA-256 integrity	100%	100%	100%
EER vs. centralised	Exact	Exact	Exact

Node failure resilience — (3, 5)-Shamir:

Failed nodes	Available	Reconstruction
0	5	100%
1	4	100%
2	3	100%
3	2	0%
4	1	0%

Sharp threshold at $t=3$: below threshold gives *exactly* 0% reconstruction — no partial leakage, no degraded reconstruction. Theorem 2 confirmed.

Enrollment and Verification Latency

Enrollment latency breakdown (ms):

Phase	Ganache	Fabric
Biometric processing (IBFV)	7.2	7.2
Serialization	0.2	0.4
Shamir split (3, 5)	0.7	1.7
IPFS upload ($\times 5$ shares)	116.4	110.4
Blockchain TX (CID commit)	245.7	693.6
Total	363 ms	806 ms

Verification latency breakdown (ms):

Phase	Ganache	Fabric
Blockchain CID lookup	66.5	53.0
IPFS share retrieval ($t=3$)	3.3	2.7
Shamir reconstruct	0.3	0.4
IBFV biometric verify	13.0	13.0
Total	83 ms	70 ms

Crypto overhead < 2 ms; all overhead is network I/O.

Sequential throughput:

Platform	Store TPS	Lookup TPS
Ganache	6.1	18.2
Fabric (1 worker)	1.5	17.2
Fabric (4 workers)	2.1	48.1

National ID system (10M users, 5 years): requires $\approx$ **0.06 TPS** sustained — both platforms comfortably exceed this. **Storage per user:**

(t, n)	Vault	IPFS	On-chain (blockchain)
(3, 5)	16.6 KB	166 KB	0.5 KB
(4, 7)	16.6 KB	232 KB	0.5 KB

Scalability:

- ▶ Verification latency is **O(1)** in enrolled population size (key-based CID lookup, no scan)
- ▶ 1M users $\rightarrow$ IPFS: 162 GB; blockchain: 488 MB
- ▶ Fabric (4 workers): read TPS scales $2.9\times$ with concurrency

- ✓ **Irreversibility:** Recovering minutia q_i from $P_{i,x}$ requires ECDLP (2^{128} ops). Iris key brute-force: 2^{514} SHA-256 evaluations. Bonus x-coords: SHA-256 preimage resistance without κ .
- ✓ **Unlinkability:** Independent polynomial, chaff, iris key per enrollment. Diverse f per application: $D_{\text{sys}} \leq 0.03$.
- ✓ **Revocability:** Re-enroll with new (f', k', K') : new polynomial domain, new iris helper data. Old vault: zero advantage (Theorem 2).

Honest disclosure — known limitation:

Expected offline brute-force: $\frac{\binom{124}{11}}{\binom{24}{11}} \approx 6.8 \times 10^8$ attempts at $10^6/\text{sec} \approx 680$ seconds.

This is a *known property* of all fuzzy vault schemes (Tams 2013).

Mitigations (three layers):

1. Store public key S in HSM — vault alone cannot accept without S
2. Increase k to 15: $\approx 3 \times 10^6$ years at $10^6/\text{sec}$
3. Argon2 key-stretching over polynomial coefficients

Note: even after vault cracking, biometric recovery still requires ECDLP (2^{128}) — irreversibility holds.

Conclusion

Method	Field	Best EER (%)	Multimodal	Zero-Penalty	ISO 24745	Distributed
Clancy (2003)	GF(2^{16})	20–30	No	—	Partial	No
Nandakumar (2007)	GF(2^{16})	2.00	No	—	Partial	No
Nandakumar & Jain (2008)	GF(2^{16})	1.50	✓	✗	Partial	No
Moon (2010)	GF(p)	3.00	✓	✗	Partial	No
Sandhya & Prasad (2017)	GF(p)	0.880	No	—	Partial	No
Maurya (2025)	ECC P256	0.005	No	—	Full	No
IBFV (S1)	ECC P256	0.000	✓	✓	Full	✓
IBFV (S2)	ECC P256	2.242	✓	✓	Full	✓

- ▶ Only multimodal ECC vault with a **formal zero-penalty proof**
- ▶ Only system satisfying all three ISO/IEC 24745 properties while providing multimodal fusion
- ▶ Only system combining feature-level fusion with threshold-distributed storage

What was built:

- ▶ Iris-derived bonus points injected as guaranteed genuine polynomial evaluations inside an ECC fuzzy vault
- ▶ Virtual x-coordinates derived via SHA-256 PRF from iris key hash: computationally indistinguishable from chaff without the key
- ▶ Smart decoder reduces search space by $5.5\times$ when iris succeeds
- ▶ Clean fallback to unimodal when iris fails

What was proven:

- ▶ **Theorem 1:** $\Pr[\text{Accept}_{\text{IBFV}}] \geq \Pr[\text{Accept}_{\text{Uni}}]$ — always
- ▶ First multimodal fuzzy vault with a formal zero-penalty guarantee

What the numbers say:

- ▶ **Up to 61.3% relative EER reduction** on the hardest FVC databases
- ▶ Zero violations of Theorem 1 in all 144 Setup-2 configurations
- ▶ Outperforms all four competing multimodal architectures on every database

What was built:

- ▶ (t, n) -Shamir Secret Sharing over GF(257) applied to the IBFV vault
- ▶ Shares stored on IPFS (content-addressed, decentralized)
- ▶ CID registry on both Ethereum/Ganache and Hyperledger Fabric
- ▶ Complete enrollment and verification protocols

What was proven and measured:

- ▶ Theorems 2 and 3: confidentiality and exact reconstruction
- ▶ **Zero bit errors** across 30,000 reconstruction trials
- ▶ Enrollment < 806 ms; verification < 83 ms; 0.5 KB on-chain
- ▶ ISO/IEC 24745: all three properties satisfied

Future work:

- ▶ Adaptive K from real-time iris quality estimation
- ▶ Co-acquired multimodal dataset (non-chimeric evaluation)
- ▶ Verifiable Secret Sharing (Pedersen 1991) for Byzantine fault tolerance
- ▶ Production deployment on Polygon / Arbitrum (Layer-2)
- ▶ Extension to additional modalities: face, palm, periocular

Thank You

Questions Welcome

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Backup Slides

For panel Q&A — not part of main presentation

For Architectures B, C, D the iris is a mandatory decryption key:

- ▶ **Arch B:** polynomial masked with iris key — without key, polynomial cannot be unmasked
- ▶ **Arch C:** iris key controls PRNG for $20\times$ chaff inflation — without key, $20\times$ more subsets to search
- ▶ **Arch D:** vault encrypted with AES-256-GCM using iris key — without key, vault is ciphertext

The floor calculation:

Iris GAR $\approx 62\% \Rightarrow$ FRR $\geq 38\%$ regardless of fingerprint quality.

EER $\approx 38\%/2 = 19\%$ regardless of parameters. **No tuning escapes this:** Lowering the match threshold raises

FAR to meet FRR at 19%. The EER floor is 19% no matter what. Arch A is somewhat better (33% of queries not blocked) but still degrades unimodal by forcing half the genuine set through the iris gate.

B2: Iris Key Recovery — GAR/FAR Detail

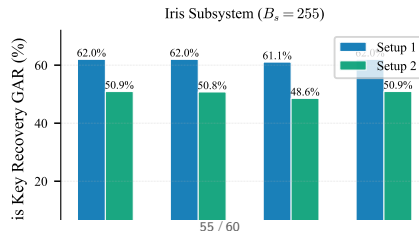
	DB1	DB2	DB3	DB4
Setup 1 GAR	62.3%	62.3%	61.1%	62.3%
Setup 2 GAR	50.9%	50.8%	48.6%	50.9%
Impostor FAR	0.10%	0.10%	0.11%	0.10%

Why Setup 2 GAR drops to ~50%:

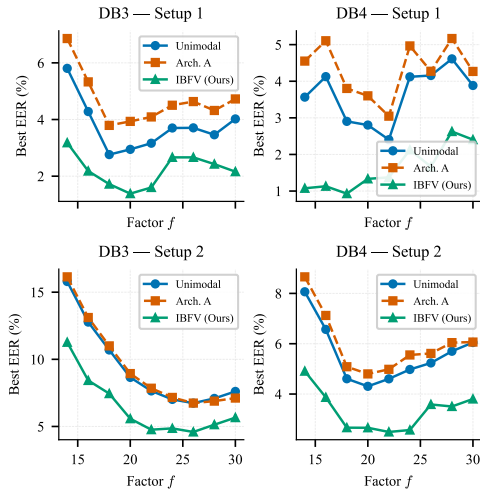
FVC2002 impressions 3–8 were captured with deliberate rotation and displacement. Cross-impression FHD increases → more blocks fail majority vote. But IBFV still achieves zero violations in Setup 2. **Impostor FAR**

≈0.10%:

≈ 2 impostors per 1,980 iris comparisons happen to recover the key at $B_s = 255$. These 2 receive K bonus points → explains the 31 marginal EER elevations (≤ 0.07 pp) on easy DBs in Setup 1 (DB1, DB2 already at 0% EER unimodal).

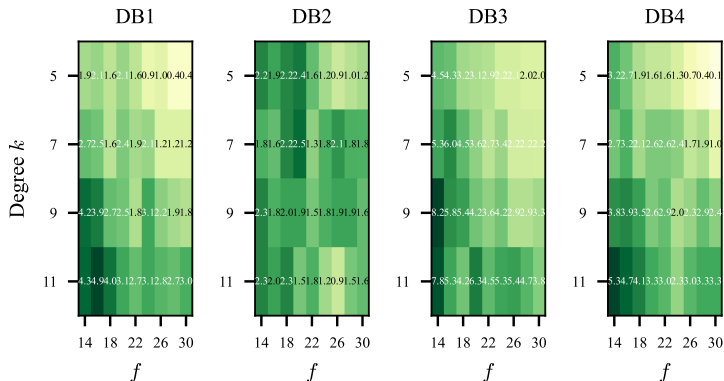


B3: EER vs. Quantization Factor



IBFV (solid) lies consistently below unimodal (dashed) at every quantization factor f across all four databases in both setups. Improvement is largest at moderate f where fingerprint matching is hardest.

Δ EER (Unimodal – IBFV) in pp — Setup 2



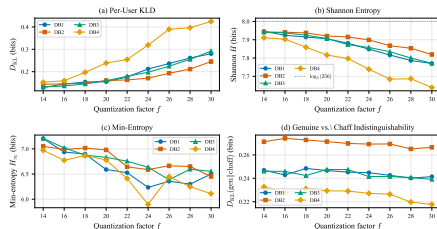
All 144 values ≤ 0 : IBFV minus Unimodal EER is non-positive in every single Setup-2 configuration. Zero violations of Theorem

1. Largest gains on DB3/DB4 at moderate f .

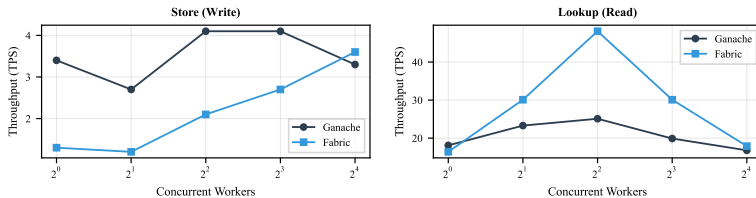
Entropy metrics ($f=24$, $k=7$, unimodal baseline):

Metric	DB1	DB2	DB3	DB4
D_{KL}^{user} (bits)	0.21	0.17	0.20	0.32
D_{KL}^{GC} (bits)	0.24	0.27	0.24	0.23
Shannon H (bits)	7.85	7.90	7.86	7.74
Min-entropy H_{∞}	6.23	6.59	6.64	5.90

- $H_{\infty} \geq 5.9$ bits: each coordinate requires ≥ 60 random guesses
- Low D_{KL} : genuine and chaff points are near-uniformly distributed — attacker cannot distinguish them
- IBFV vs unimodal entropy difference: < 0.001 bits — bonus points do not degrade the vault's randomness



B6: Concurrent Throughput Scaling



Workers	Store TPS		Lookup TPS	
	Ganache	Fabric	Ganache	Fabric
1	3.4	1.3	18.1	16.4
2	2.7	1.2	23.3	30.1
4	4.1	2.1	25.1	48.1
8	4.1	2.7	19.9	30.1

Fabric peaks at 4 workers for reads ($2.9\times$ single-client). Biometric systems: read:write ratio is typically 1000:1 in production.

Read scaling is the critical bottleneck — Fabric handles it.

B7: Blockchain Comparison with Prior Systems

System	Template Protection	Vault Split	Formal Proofs	On-chain (KB)	EER (%)
Delgado-Mohatar	Cancelable	None	No	Several	N/R
Goel	Raw biometric	None	No	Full	N/R
Lee (BDAS)	Fragmented	None	No	Distrib.	N/R
Sharma (2024)	Cancelable	None	No	Full	~8.0
Yun (Bio-Rollup)	Hash only	None	No	Batch	N/R
D-IBFV Ganache	ECC vault + SSS	(t, n) -Shamir	Yes	0.5	0.000
D-IBFV Fabric	ECC vault + SSS	(t, n) -Shamir	Yes	0.5	0.000

Only system with vault-level Shamir splitting, formal proofs, and sub-1 KB on-chain footprint.

How IPFS stores and retrieves data:

- ▶ Data is split into blocks; each block is hashed (SHA-256)
- ▶ Blocks are organized into a **Merkle Directed Acyclic Graph (DAG)**: parent node hash depends on all child hashes
- ▶ **Content Identifier (CID)**: a multihash encoding the hash algorithm + hash of the root block. CIDv1 = base32(multicodec + multihash)
- ▶ Retrieving a CID: broadcast to network → any node that has pinned it responds → hash is re-verified locally on receipt

Pinning:

- ▶ By default IPFS garbage-collects unpinned data
- ▶ Pinning a CID means a node commits to storing it permanently
- ▶ In D-IBFV: enrollment node pins all n shares; additional nodes can be incentivized (Filecoin)

Why integrity is automatic:

Tampered share → different SHA-256 → different CID → blockchain lookup fails → retrieval is rejected. No separate integrity check needed.

Execute-Order-Validate (EOV) model:

1. **Execute:** Client submits transaction proposal to endorsing peers. Peers simulate chaincode execution, sign read/write set.
2. **Order:** Client collects endorsements, submits to Ordering Service. Raft consensus produces a block containing the transaction.
3. **Validate:** All peers validate endorsement policy and MVCC consistency. Valid transactions update the ledger state.

D-IBFV Fabric configuration:

- ▶ Go chaincode: CID enrollment, lookup, revocation
- ▶ Endorsement policy: 1-of-1 (single org; extendable)
- ▶ Raft ordering: 3-node crash fault tolerant, 0.5-second block timeout
- ▶ Channel isolation: D-IBFV transactions on dedicated channel

Why EOV matters for biometric systems: No global ordering required before execution → parallel endorsement by multiple peers → higher throughput than Ethereum's sequential execution.

Choosing (t, n) for deployment:

(t, n)	Fault tol.	Split (ms)	Recon (ms)	IPFS storage	Security
(2, 3)	1 node	0.3	0.2	99 KB	Min
(3, 5)	2 nodes	0.7	0.2	166 KB	Moderate
(4, 7)	3 nodes	1.3	0.2	232 KB	Strong
(5, 9)	4 nodes	2.1	0.3	299 KB	Strongest

- Reconstruction time is **always** 0.2–0.3 ms regardless of (t, n) — Lagrange interpolation is fast
- Splitting time scales linearly with n
- Storage scales linearly with n
- Fault tolerance = $n - t$ nodes can fail
- (3, 5) is the recommended default: tolerates 2 simultaneous failures with modest storage overhead

